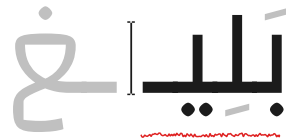


Cairo University
Faculty of Engineering
Department of Computer Engineering



Baligh

Arabic Writing Assistant

A Graduation Project Report Submitted

to

Faculty of Engineering, Cairo University

in Partial Fulfillment of the requirements of the degree

of

Bachelor of Science in Computer Engineering.

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Abstract

Arabic digital writing still suffers from a shortage of integrated tools that can correct Modern Standard Arabic text accurately, respond in real time, and explain their feedback in a way that helps users learn. Existing tools often focus on surface spelling fixes or black-box suggestions without exposing the grammatical reasoning behind the correction. Baligh addresses this problem by providing an Arabic writing assistant that combines grammatical error detection, grammatical error correction, spelling-aware editing, next-word suggestion, and explanatory feedback in a single user-facing system. The project aims to support students, educators, and Arabic content writers by improving writing quality while also making the correction process more understandable and educational.

The implemented system follows a modular pipeline. A preprocessing layer performs normalization, word-boundary handling, segmentation, and morphological analysis and disambiguation. On top of that, Baligh integrates a grammatical error detection engine that fuses rule-based, lexicon-based, and machine-learning detectors, then passes its findings to correction components built around ontology-guided reasoning, dictionary lookup, and text-editing models. In parallel, a next-word suggestion module supports both next-word prediction and word auto-completion. The final system highlights issues, proposes corrections, returns ranked suggestions, and presents grammar-oriented explanations to the user in an interactive workflow.

The project outputs include the preprocessing pipeline, grammatical error detection and correction services, the next-word suggestion subsystem, and a working user application. Testing and validation relied on automated unit and integration tests together with evaluation on established Arabic linguistic resources, including QALB14, QALB15, and ZAEBUC. Overall, Baligh demonstrates that a modular and explainable Arabic writing assistant can combine linguistic processing and machine learning into a practical platform for improving Arabic writing quality.

الملخص

لا تزال الكتابة الرقمية باللغة العربية تعاني من نقص الأدوات المتكاملة التي تستطيع تصحيح نصوص العربية الفصحى بدقة، والعمل بزمان استجابة مناسب، وتقديم تفسير واضح يساعد المستخدم على التعلم بدل الاكتفاء بإظهار التصحيح فقط. فكثير من الأدوات المتاحة تركز على الأخطاء الإملائية السطحية أو تقدم اقتراحات غير مفسرة. يعالج مشروع بليغ هذه المشكلة من خلال بناء مساعد كتابة عربي يجمع بين اكتشاف الأخطاء النحوية، وتصحيحها، ومعالجة الأخطاء الإملائية، واقتراح الكلمة التالية، وتقديم تفسير مبسط للقاعدة أو سبب الخطأ داخل نظام واحد موجه للمستخدم. ويهدف المشروع إلى خدمة الطلاب والمعلمين وصناع المحتوى العربي عبر تحسين جودة الكتابة وجعل عملية التصحيح أكثر فائدة من الناحية التعليمية.

يعتمد النظام المنفذ على بنية معيارية تبدأ بطبقة للمعالجة المسبقة تشمل التطبيع، وتحديد حدود الكلمات، والتجزئة، ثم التحليل الصرفي وفك الالتباس. بعد ذلك يدمج بليغ محركاً لاكتشاف الأخطاء النحوية يعتمد على المزج بين قواعد لغوية، وكشافات معجمية، ونماذج تعلم آلي، ثم يمرر النتائج إلى وحدات التصحيح المبنية على الاستدلال المعتمد على الأنطولوجيا، والبحث المعجمي، ونماذج التحرير النصي. وبالتوازي مع ذلك، يوفر النظام وحدة لاقتراح الكلمة التالية تشمل التنبؤ بالكلمة التالية والإكمال التلقائي للكلمة الجارية كتابتها. وتعرض المنصة النهائية الأخطاء والتصحيحات والاقتراحات والتفسيرات اللغوية بصورة تفاعلية.

تشمل مخرجات المشروع خط معالجة مسبقة متكامل، وخدمات لاكتشاف الأخطاء وتصحيحها، ووحدة اقتراح الكلمات، وتطبيقاً عملياً للمستخدم. واعتمد التحقق من النظام على اختبارات آلية للوحدات والتكامل، إضافة إلى تقييم مبني على موارد عربية معروفة مثل QALB14 و QALB15 و ZAEUC. وتُظهر المحصلة أن بليغ يقدم نموذجاً عملياً لمساعد كتابة عربي قابل للتفسير يجمع بين المعالجة اللغوية والتعلم الآلي لخدمة الكتابة العربية بصورة أكثر جودة وفائدة.

Acknowledgment

Write your acknowledgment and dedication here.

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List of Abbreviations

Put abbreviations in alphabetical order.

Abbreviation	Meaning
AI	Artificial Intelligence
GEC	Grammatical Error Correction
GED	Grammatical Error Detection
MSA	Modern Standard Arabic
NLP	Natural Language Processing
NWP	Next-Word Prediction

List of Symbols

Put symbols in alphabetical order. Greek symbols should come first, followed by English symbols.

Symbol	Meaning
α	Learning rate or weighting coefficient
P	Probability
T	Time or sequence length

Contacts

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Chapter 1: Introduction

1.1. Project Overview

Briefly introduce Baligh and the overall graduation project.

1.2. Individual Role Overview

Summarize the student's role and responsibilities in the team.

1.3. Document Organization

Briefly describe the organization of the individual report.

Chapter 2: Personal Role and Responsibilities

2.1. Team Responsibilities

Describe coordination, research, implementation, testing, documentation, communication, or leadership responsibilities.

2.2. Assigned Tasks

List the tasks assigned to this student and how they contributed to the project objectives.

2.3. Timeline and Deliverables

Summarize major milestones and deliverables.

Chapter 3: Knowledge Acquisition and Technical Preparation

3.1. Investigated Papers

Summarize the papers/materials studied and how they influenced the student's work.

3.2. Self-Learning Materials

List courses, tutorials, documentation, and tools studied during the project.

3.3. Relevance to the Project

Explain how the acquired knowledge supported the implemented contribution.

Chapter 4: Individual Contributions

4.1. Contribution 1

Describe the first major contribution, implementation details, design decisions, and results.

4.2. Contribution 2

Describe the second major contribution. Add figures, tables, and code snippets where useful.

Sample code snippet placeholder

```
def explain_correction(sentence: str) -> str:
    """Return a human-readable explanation for a correction."""
    return "Replace this placeholder with actual project code."
```

4.3. Testing and Validation

Explain how the individual contribution was tested and validated.

Chapter 5: Challenges, Lessons Learned, and Future Work

5.1. Faced Challenges

Discuss technical and personal challenges.

5.2. Lessons Learned

Discuss the most important skills and insights gained.

5.3. Future Enhancements

Mention how this contribution could be improved in future work.

References

[1] Author Name, "Paper or Book Title," *Conference or Journal Name*, Year.

[2] Author Name, "Another Reference Title," Publisher or Venue, Year.

Chapter A: Development Platforms and Tools

List programming languages, frameworks, libraries, operating systems, hardware, datasets, and development tools.

Chapter B: Use Cases

Describe system use cases, actors, preconditions, main flows, alternative flows, and expected results.

Chapter C: User Guide

This appendix should be included. Explain how to install, run, and use the final system. Add screenshots or step-by-step instructions when available.

Chapter D: Code Documentation

Document important modules, classes, APIs, configuration files, and code structure.

Chapter E: Feasibility Study

This appendix should be included. Add market, technical, operational, and financial feasibility details.